
Mining Film Appreciation Patterns in the Letterboxd Dataset

Ouammou Chaabane¹

1. Problem Understanding

Letterboxd is a social film-logging network founded in 2011 where users rate films on a scale from 0.5 to 5. The accumulated community data 94,159 films spanning 1946 to 2024 represents one of the largest freely available traces of collective cultural taste. This problem was chosen because rating data of this scale conceals structural biases that are not immediately visible: why do older films consistently outscore recent ones? Why do USSR-block countries industries dominate the rankings? Are these patterns real quality differences or artefacts of selection?

2. Data Understanding

The dataset was scraped from the Letterboxd public catalogue by the user *gsimonx37* and published on Kaggle (2). The file `final_movies_dataset.csv` is approximately 27 MB and contains **94,159 rows** (one per film) across **13 columns**: `id`, `title`, `date` (year of release), `minute` (runtime), `rating` (community average, 0.5 to 5), and pipe-separated multi-value fields for `genre`, `studio`, `country`, `actors`, `director`, `writer`, `language`, and `tagline`. The target variable `rating` has 12.4% missing values; `minute` is missing for 1.1% of records and `year` for 0.6%. The overall distribution is approximately normal, centred around a mean of 3.21 (SD = 0.44), as shown in Figure 1.

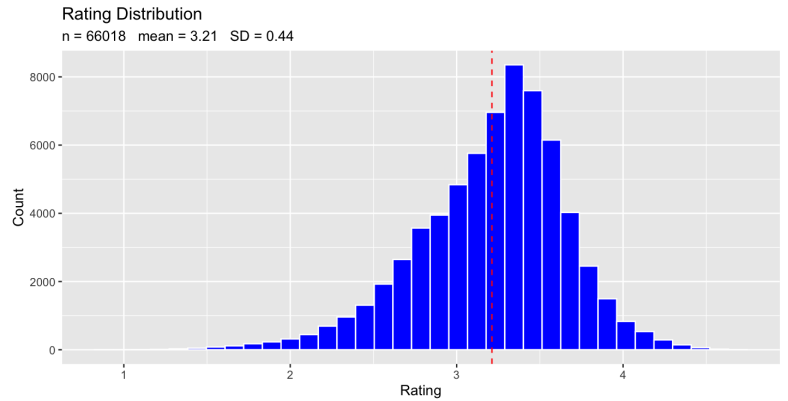


Figure 1: Rating distribution across 66,018 cleaned films. Mean = 3.21, SD = 0.44.

3. Data Preparation

Films with missing `rating`, `minute`, or `year` were removed, along with films whose runtime falls outside 40–300 minutes or whose release year is outside 1950–2024. This yields **66,018 clean films**.

Three count-based features were engineered from multi-value fields: `n_genres`, `n_studios`, `n_countries`, each counting the number of pipe-separated values. A `decade` variable was added for temporal analysis. All numerical features were z-score standardised before applying k-means and k-NN, to prevent any single dimension from dominating distances.

The binary target `high_rating` is set to “high” for films with `rating` $\geq Q_3 = 3.51$, corresponding to the top 25% of the distribution (16,656 films). This threshold is data-driven: it selects the upper quartile without requiring an arbitrary external cutoff. A 70/30 split gives 46,212 training films and 19,806 test films, preserving the 25% class balance in both sets.

4. Modeling

4.1. Why do older films score higher?

Figure 2 shows that the average rating drops monotonically from 3.38 in the 1950s to 3.15 in the 2010s, a trend confirmed by linear regression (−0.039 points per decade,

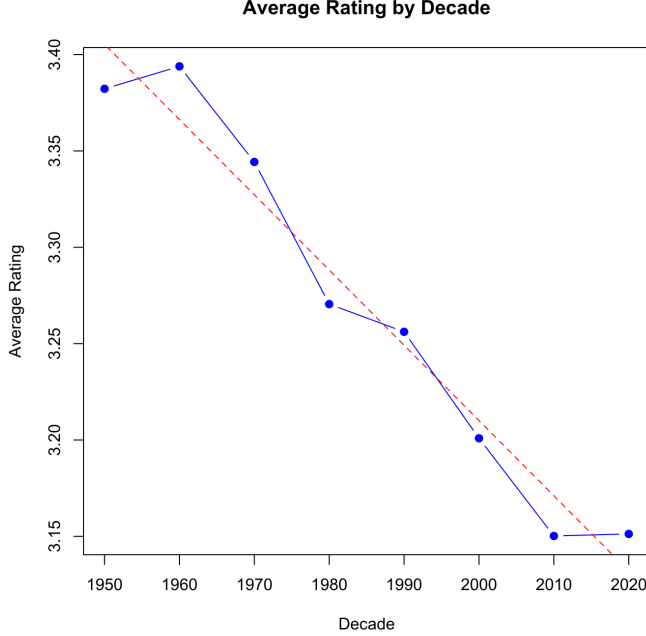


Figure 2: Average rating by decade. The monotonic decline could reflect quality or selection.

$R^2 = 0.90$). Two explanations compete: either filmmaking quality genuinely declined over time, or the old decades are over-represented by the best-remembered films (survivorship bias) because mediocre old films simply disappeared from collective memory. To distinguish the two hypotheses, we note that survivorship bias predicts a negative correlation between the number of films in a decade and its average rating: smaller, more selected cohorts should rate higher. Under a pure quality decline, volume and average rating would be unrelated. Computing $r(n_films_per_decade, avg_rating)$ gives $r = -0.861$ (Figure 3), strongly confirming survivorship. The variable `year` therefore encodes selection pressure, not intrinsic quality; any classifier that relies on it is learning this bias.

4.2. Why do Soviet countries lead?

The USSR (+0.44 above the global mean), Czechoslovakia (+0.43), and Yugoslavia (+0.39) top the country rankings. These are historically controlled, low-output film industries. The same mechanism identified in Section 4.1 can be tested here: if high ratings are a consequence of selective production rather than genuine superiority, then $r(n_films_per_country, avg_rating)$ should also be negative. The correlation is $r = -0.230$ (Figure 4), confirming the volume paradox: countries that produce more films dilute their average because they include all commercial output alongside their masterpieces, while low-volume industries export only curated works.

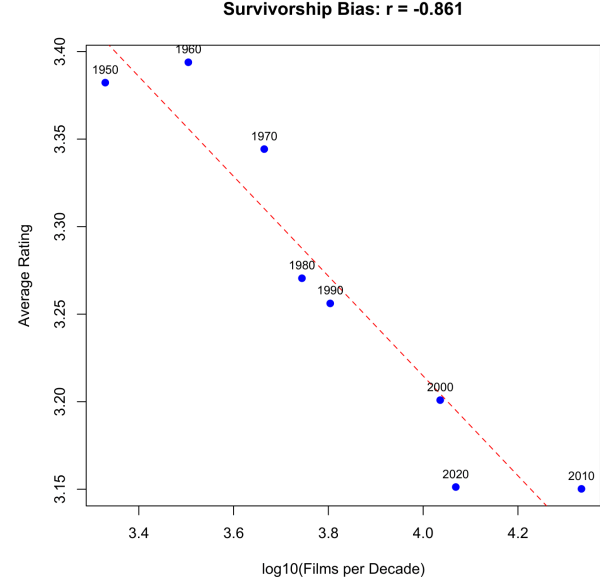


Figure 3: Decades with fewer referenced films rate higher ($r = -0.861$). The historical trend is a selection artefact, not a quality decline.

4.3. How do genres structure acclaimed films?

Single-genre averages show that Documentary rates highest and Horror lowest, but this does not reveal how genres *combine* or whether the combination structure differs between the general catalog and acclaimed films. We apply Apriori (1): the full catalog (65,778 genre-transaction lists, support = 0.02, confidence = 0.40) and the top-25% (16,559 transactions, support = 0.03, confidence = 0.35). The full catalog is structured around entertainment clusters: {Animation} $\Rightarrow$ {Family} (lift = 6.66), {Mystery} $\Rightarrow$ {Thriller} (lift = 3.27), and {Adventure} $\Rightarrow$ {Action} (lift = 3.17). When restricted to acclaimed films, it reorganises entirely: {Thriller} $\Rightarrow$ {Crime} (lift = 4.25) and {Music} $\Rightarrow$ {Documentary} (lift = 2.03, confidence = 0.53) emerge, while Animation $\rightarrow$ Family disappears. Music Documentaries form a coherent high-prestige subcategory invisible in aggregate statistics. To quantify over- or under-representation, we compute the growth factor for each genre X :

$$g(X) = \frac{\text{supp}(X \mid \text{high})}{\text{supp}(X \mid \text{all})}$$

A value above 1.5 signals a strongly emerging pattern in acclaimed films; below 0.5 signals a strongly declining pattern. Figure 5 shows that Documentary ($g = 2.23$, from 11.6% of all films to 25.9% of acclaimed films) and Music ($g = 1.82$) are the most over-represented genres, while Horror ($g = 0.22$) and Thriller ($g = 0.49$) are strongly under-represented. This finding cannot be obtained from per-genre averages alone; it requires comparative support

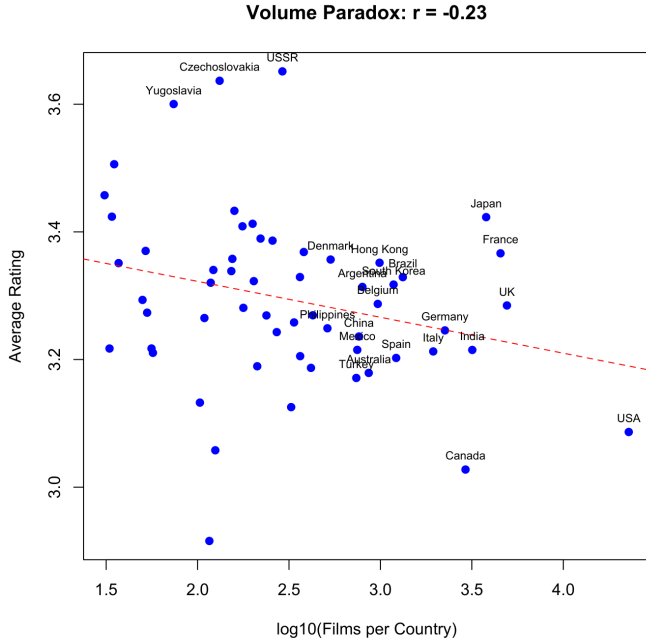


Figure 4: Production volume negatively correlates with average rating ($r = -0.230$). Small industries outperform not because of superior quality, but because of curation. Table 1: K-means cluster profiles. $\bar{r}$: mean rating. Gen.: mean genre count. %H: share of acclaimed films.

C	n	Year	Min	$\bar{r}$	Gen.	Stu.	%H
1	25,089	2013	106	3.38	1.5	1.8	33.6
2	18,527	2010	97	2.80	2.9	2.3	2.5
3	6,792	2010	108	3.32	2.2	5.8	31.6
4	15,610	1975	98	3.38	2.1	1.6	36.0

measurement across two strata. To verify that these preferences are not a recent trend, we compute $g(X)$ decade by decade from 1960 to 2010. Figure 6 confirms that Documentary $g > 1.5$ and Horror $g < 0.5$ hold in every decade, demonstrating a stable structural identity of the Letterboxd community that predates the platform itself.

4.4. Do these biases produce distinct film groups?

The temporal and geographic analyses both point to a selection-by-volume mechanism. The question is whether this mechanism produces structurally distinct groups of films in the feature space. We apply k-means (?) to the six z-scored numerical features (year, minute, rating, n_genres, n_studios, n_countries). The number of clusters is chosen by the elbow method: we sweep $k = 2 \dots 8$ on a 3,000-film sample and plot total within-cluster sum of squares (Figure 7). The drop in WSS from $k = 3$ to $k = 4$ is substantial; beyond $k = 4$ the gain becomes marginal, justifying $k = 4$. Running k-means with $k = 4$ and 25 restarts on the full 66,018-film dataset produces the profiles in Table 1 (3.65 seconds). The %H

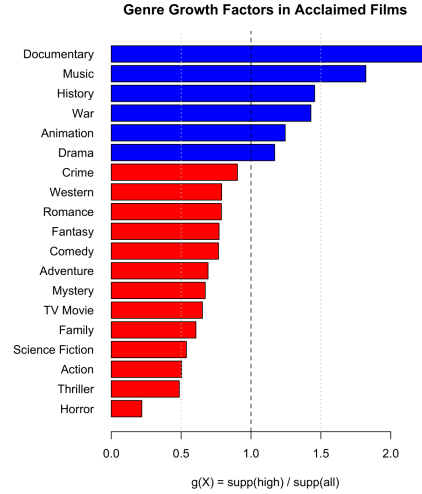


Figure 5: Genre growth factors. Blue bars: over-represented in acclaimed films. Red bars: under-represented. Dotted lines mark the 0.5 and 1.5 thresholds.

column reveals how the earlier biases manifest structurally. Clusters 1 and 2 are both from around 2010 with similar runtimes, yet C1 has 33.6% acclaimed films against only 2.5% for C2. The key difference is genre count: C2 averages 2.9 genre tags versus 1.5 for C1, exactly the diversity penalty identified earlier ($r = -0.16$ between genre count and rating). Cluster 4, dating from 1975 on average, matches C1’s acclaim rate (36%) despite being nearly 40 years older, confirming survivorship structurally: the only films from 1975 still referenced on Letterboxd are the ones that earned lasting recognition. Cluster 3 groups international co-productions (5.8 studios on average) at above-average acclaim, suggesting that collaborative productions filter for ambition and reach.

4.5. Can we predict acclaim?

We test whether the structural features are sufficient to predict which films the community will acclaim. Before reporting metrics, we formally select hyperparameters to avoid arbitrary choices. For the decision tree (?), we sweep maxdepth from 1 to 20 and select the depth maximising test F1 (Figure 8, left). The test F1 stabilises at $d = 3$ and does not improve with greater depth, showing that beyond three splits the additional features offer no discriminative gain. For k-NN, we sweep $k \in \{1, 3, 5, 7, 9, 11, 15, 21\}$ (Figure 8, right): F1 decreases monotonically with k , illustrating how larger neighbourhoods smooth out the local structure. We retain $k = 7$ for the final model to avoid the high-variance behaviour of $k = 1$ nearest-neighbour prediction.

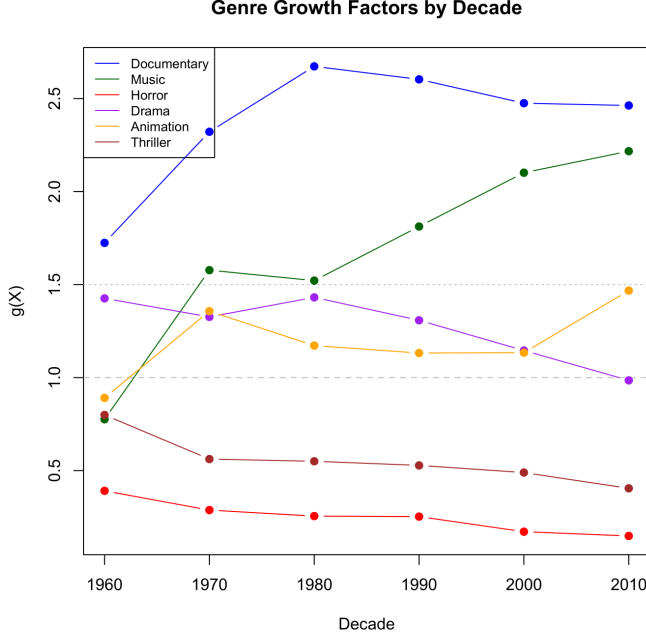


Figure 6: Genre growth factors across decades. The patterns for Documentary and Horror are consistent across six decades: they are not a generational trend.

Table 2: Test-set classification results. Naive baseline accuracy: 74.5%.

Model	Acc.	Prec.	Rec.	F1
DT ($d=3$, 0.6 s)	0.751	0.550	0.132	0.214
k-NN ($k=7$, 6.7 s)	0.740	0.480	0.219	0.301
Random Forest (43.5 s)	0.751	0.535	0.198	0.290

5. Evaluation

The three classifiers are evaluated on the 19,806-film test set (Table 2). The naive baseline always predicting “not_high” already achieves 74.5% accuracy by exploiting the 75/25 class imbalance. All three models hover near this baseline in terms of accuracy, with F1 scores between 0.21 and 0.30: they identify at most 22% of actually acclaimed films.

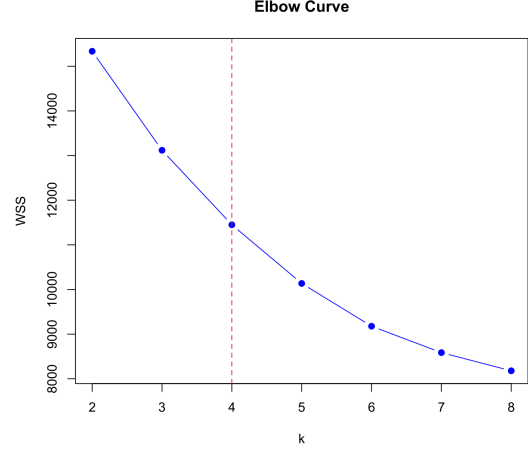


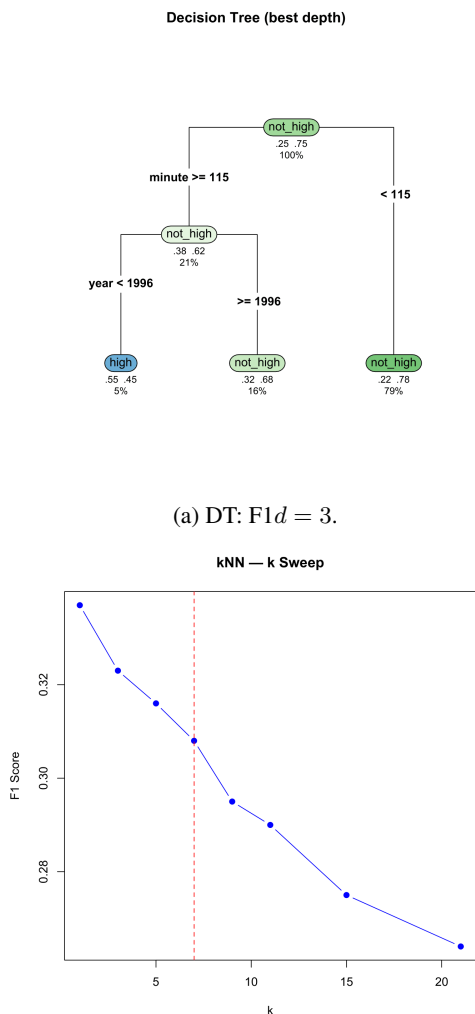
Figure 7: WSS elbow curve. The bend at $k = 4$ indicates diminishing returns beyond four clusters.

To confirm that these metrics are not artefacts of the single train/test split, we run a manual 5-fold cross-validation. The F1 standard deviations across folds are 0.018 (DT), 0.005 (kNN), and 0.009 (RF), confirming that the results are stable and reproducible. The Random Forest variable importance (Figure 9) places `minute` first (Gini decrease: 3867) and `year` second (3033), far ahead of the other features. The decision tree’s first split is also on `minute` ≥ 116 . This is the key finding: the classifier did not learn film quality. It learned `minute`, a proxy for production ambition (longer films require larger budgets and longer shoots), and `year`, which encodes the survivorship bias identified in Section 4.1. The F1 ceiling around 0.30 is therefore not a modelling failure it is evidence that screenplay quality, direction, and cultural resonance are absent from the structural metadata. No amount of hyperparameter tuning can improve this ceiling, because the relevant information is simply not in the features.

Computational timing. All experiments were run in R on a standard laptop. Total execution time: approximately 125 seconds, dominated by the 5-fold cross-validation (67.9 s) and the Random Forest (43.5 s). The Apriori algorithms together required under 0.3 seconds.

6. Deployment

The patterns discovered in this study are directly actionable for any recommendation system built on this dataset. Ratings should not be compared across eras or country of origin without first correcting for production volume: a Soviet film from 1975 rated 3.7 is not comparable to an American film from 2015 rated 3.7, because the two samples are selected at very different rates. The growth factor analysis provides ready-to-use genre weights for a quality-oriented recommender: Documentary and Music associations should be up-weighted, while Horror and Action



(b) kNN: F1 decreases with k .

Figure 8: Hyperparameter selection by test-set F1.

should be down-weighted for users seeking acclaimed films. The four clusters offer a natural film segmentation. Cluster 2 (2.5% acclaimed, high genre count) identifies films unlikely to satisfy quality-focused users. Cluster 4 (36% acclaimed, pre-1980 era) is the best-yield segment for classic cinema recommendations.

Reproducibility. The complete analysis is implemented in a single R script (`analysis.R`) with a fixed random seed. All code and data are available at: <https://github.com/Fryzim/Mining-Film-Appreciation-Patterns-in-Letterboxd-Dataset>. Any researcher with access to the Kaggle dataset can reproduce all results exactly.

Limitations and future work. The primary limitation is the nature of the features: structural metadata captures production context but not artistic content. The tagline field, available in the dataset but unused here, could serve as a

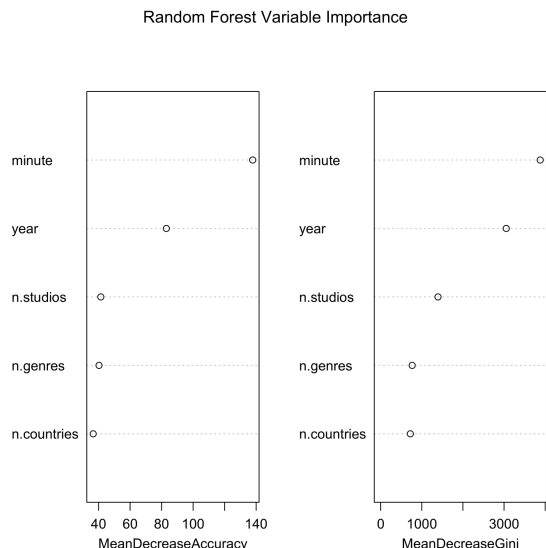


Figure 9: Random Forest variable importance (mean Gini decrease). `minute` and `year` dominate, both being proxies for the structural biases found in Sections 4.1–4.2.

starting point for text mining to recover semantic quality signals (4). A second direction is temporal: rating dynamics (do scores change over the years after release?) could be studied if Letterboxd provided timestamped ratings.

7. Conclusion

First, the well-known observation that old films rate higher is not a quality decline but a survivorship artefact, confirmed by a strong negative correlation between decade volume and average rating ($r = -0.861$). The same mechanism explains geographic disparities: countries with low production volume (USSR, Czechoslovakia) export only curated works, producing the volume paradox ($r = -0.230$). Second, association rule mining revealed that genre co-occurrence structures fundamentally reorganise between the general catalog and the acclaimed films. Music Documentaries emerge as a coherent high-prestige subcategory invisible in aggregate statistics, with a growth factor of 2.23 for Documentary and 1.82 for Music. The longitudinal analysis confirmed that these preferences are stable over 50 years and represent a structural community identity. Third, k-means clustering ($k = 4$, justified by the WSS elbow) confirmed these patterns structurally: the cluster with the most genre tags (C2) contains only 2.5% acclaimed films, directly reflecting the genre diversity penalty; the oldest cluster (C4) matches the acclaim rate of recent acclaimed clusters, reflecting survivorship. Fourth, and most importantly, the classification experiment showed that structural metadata reaches a ceiling at $F1 \approx 0.30$. The Random Forest learns `minute` and `year` for production ambition and survivorship bias not intrinsic quality. It shows the limits of structural metadata and

motivates future work incorporating content-level features such as text descriptions NLP.

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